

An Adapted Version

# Introductory Combinatorics

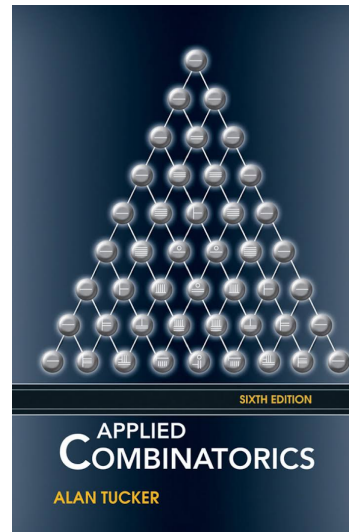
5th Edition

By Richard A. Brualdi



## Chapter - Pólya Counting

# applied combinatorics



6th Edition

By Alan Tucker

**TABLE 1** Combinations and Permutations with and without Repetition.

| <i>Type</i>       | <i>Repetition Allowed?</i> | <i>Formula</i>              |
|-------------------|----------------------------|-----------------------------|
| $r$ -permutations | No                         | $\frac{n!}{(n-r)!}$         |
| $r$ -combinations | No                         | $\frac{n!}{r!(n-r)!}$       |
| $r$ -permutations | Yes                        | $n^r$                       |
| $r$ -combinations | Yes                        | $\frac{(n+r-1)!}{r!(n-1)!}$ |

# Distributing objects into Boxes :

Distinguishable Objects and distinguishable boxes

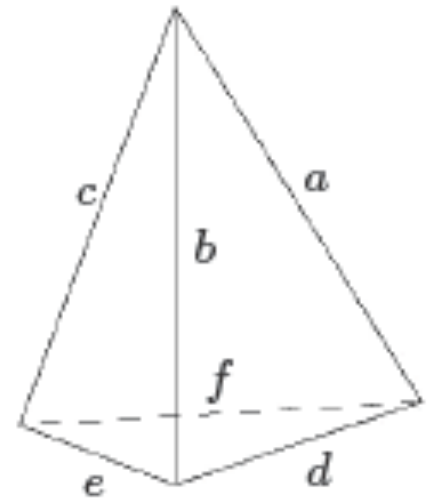
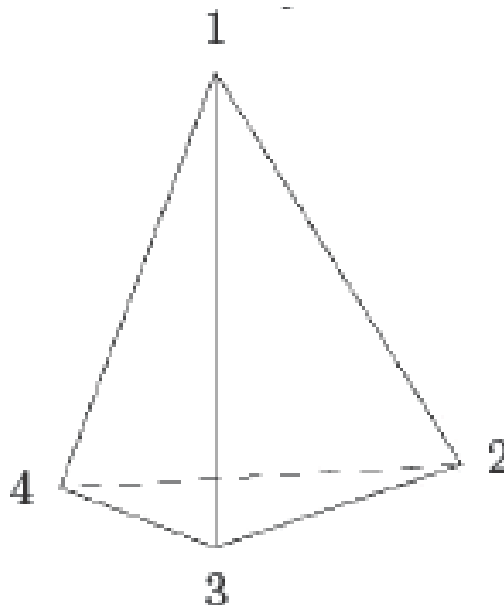
**I**n distinguishable Objects and distinguishable boxes

Distinguishable Objects and **i**n distinguishable boxes

**I**n distinguishable Objects and **i**n distinguishable boxes

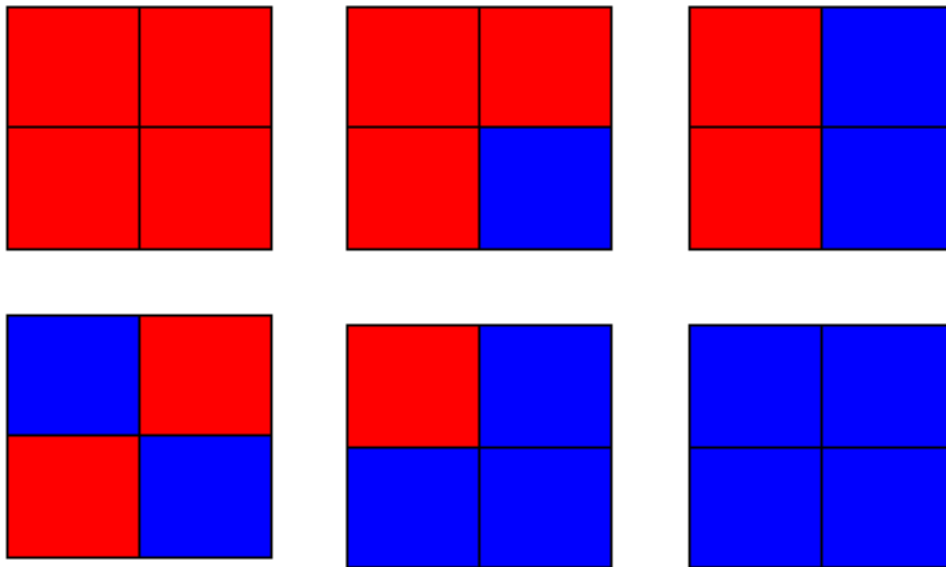
- Geometric symmetries:
  - suppose you wish to color the four corners of a **regular tetrahedron**, and you have just two colors, red and blue.
  - How many different colorings are there?

$$2^4 = 16 ?$$



- **Example 1** A disc lies in a plane. Its centre is fixed but it is free to rotate. It has been divided into  $n$  sectors of angle  $2\pi/n$ . Each sector is to be colored Red or Blue. How many different colorings are there?
- One could argue for  $2^n$ .
- On the other hand, what if we only distinguish colorings which cannot be obtained from one another by a rotation. For example if  $n = 4$  and the sectors are numbered 0,1,2,3 in clockwise order around the disc, then there are only 6 ways of coloring the disc – 4R, 4B, 3R1B, 1R3B, RRBB and RBRB.

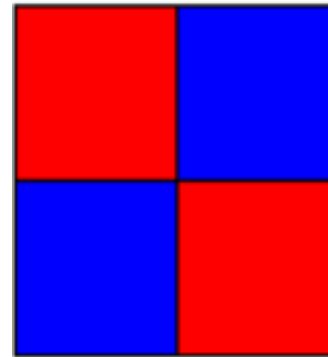
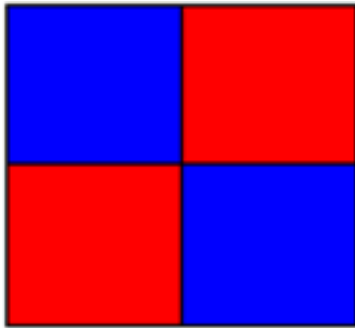
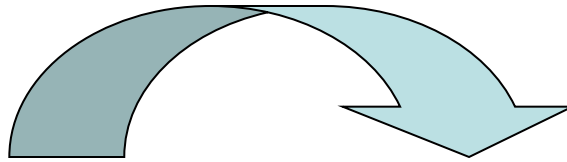
- Example 2** Now consider an  $n \times n$  “chessboard” where  $n \geq 2$ . Here we color the squares Red and Blue and two colorings are different only if one cannot be obtained from another by a rotation or a reflection. For  $n = 2$  there are 6 colorings.



$$2^{n^2} = 16$$

$$\checkmark 6$$

rotate 90°



# Six subsets of equivalent colorings vary in size

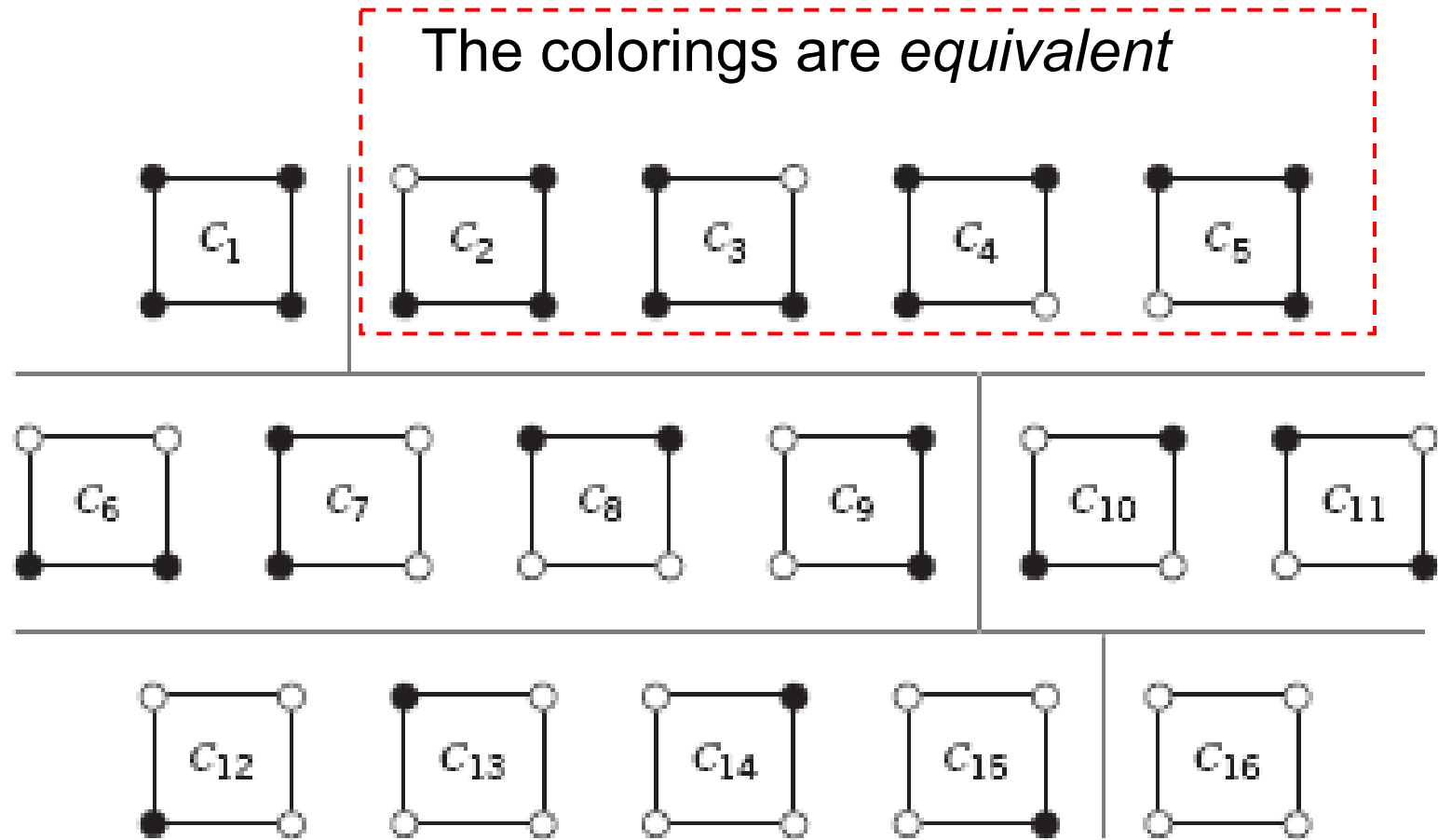


Figure 9.1

- Permutation and Symmetry groups
  - Let Each permutation  $i_1, i_2, \dots, i_n$  of  $X$  can be viewed as a one-to-one function

$$X = \{1, 2, \dots, n\}$$

$$f(1) = i_1, f(2) = i_2, \dots, f(n) = i_n$$

- We also denote this permutation by 2-by- $n$  array

$$\begin{pmatrix} 1 & 2 & \dots & n \\ i_1 & i_2 & \dots & i_n \end{pmatrix}$$

- Permutation and Symmetry groups
  - The motions we want to consider can be thought of as permutations, that is, bijections.

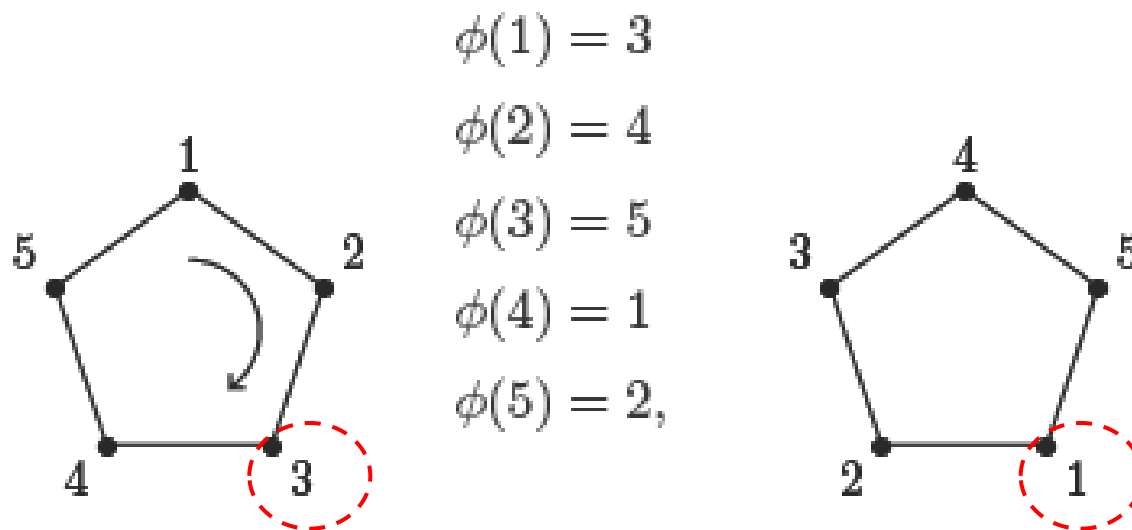


Figure 6.1.1 The rotation  $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 1 & 2 \end{pmatrix}$ .

- Example. The  $3!=6$  permutations of  $\{1, 2, 3\}$

$$\begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}$$

- We denote the set of all  $n!$  permutations of  $\{1, 2, \dots, n\}$  by  $S_n$ . For example,  $S_3$  consists of 6 permutations. Since permutations are functions, they can be combined, using **composition**

$$f = \begin{pmatrix} 1 & 2 & \dots & n \\ i_1 & i_2 & \dots & i_n \end{pmatrix}$$

$$g = \begin{pmatrix} 1 & 2 & \dots & n \\ j_1 & j_2 & \dots & j_n \end{pmatrix}$$

$$g \circ f = \begin{pmatrix} 1 & 2 & \dots & n \\ i_1 & i_2 & \dots & i_n \end{pmatrix} \circ \begin{pmatrix} 1 & 2 & \dots & n \\ j_1 & j_2 & \dots & j_n \end{pmatrix}$$

$$(g \circ f)(k) = g(f(k)) = j_{i_k}$$

Composition of functions defines a *binary operation* on  $S_n$ : if  $f$  and  $g$  are in  $S_n$ , then  $g \circ f$  is also in  $S_n$

- Example. Let  $f$  and  $g$  be the permutation in  $S_4$

$$f = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix}, g = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix}$$

$$(g \circ f)(1) = 3, (g \circ f)(2) = 4, (g \circ f)(3) = 1, (g \circ f)(4) = 2$$

$$g \circ f = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix}$$

$$f \circ g = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix}$$

- In general

$$(g \circ f) \circ h = g \circ (f \circ h) \quad \text{associative law}$$

$$f \circ g \neq g \circ f \quad \text{But not satisfy the commutative law}$$

$$f^1 = f, f^2 = f \circ f, f^3 = f \circ f \circ f, \dots, f^k = f \circ f \circ \dots \circ f \text{ (k f's)}$$

$$I = \begin{pmatrix} 1 & 2 & \dots & n \\ 1 & 2 & \dots & n \end{pmatrix} \quad \text{identity permutation: the permutation of } \{1, 2, \dots, n\} \text{ for all } I(k) = k$$

$$I \circ f = f \circ I = f$$

$$f^{-1}(k) = s \Rightarrow f(s) = k$$

- Example. Consider the permutation in  $S_6$

$$f = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 5 & 6 & 3 & 1 & 2 & 4 \end{pmatrix}$$

$$\begin{pmatrix} 5 & 6 & 3 & 1 & 2 & 4 \\ 1 & 2 & 3 & 4 & 5 & 6 \end{pmatrix} \quad \text{Interchange rows 1 and 2}$$

$$f^{-1} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 4 & 5 & 3 & 6 & 1 & 4 \end{pmatrix} \quad \text{Rearrange columns}$$

$$f \circ f^{-1} = f^{-1} \circ f = \iota$$

- A group of permutations of  $X$ , (in abbreviated form, a *permutation group*), is defined to be a nonempty subset  $G$  of permutations in  $S_n$  satisfying the following three properties
  - closure under composition
    - For all permutations  $f$  and  $g$  in  $G$ ,  $f \circ g$  is also in  $G$
  - identity
    - The identity permutation  $1$  of  $S_n$  belongs to  $G$
  - closure under inverse
    - For each permutation  $f$  in  $G$  the inverse  $f^{-1}$  is also in  $G$

- The set  $S_n$  of all permutations of  $X=\{1, 2, \dots, n\}$  is a permutation group, is also called the *symmetric group of order  $n$*

$$f \circ g = f \circ h \quad \boxed{\text{cancellation law}}$$

$$f^{-1} \circ (f \circ g) = f^{-1} \circ (f \circ h) \quad \boxed{\text{associative law}}$$

$$(f^{-1} \circ f) \circ g = (f^{-1} \circ f) \circ h$$

$$\iota \circ g = \iota \circ h$$

$$g = h$$

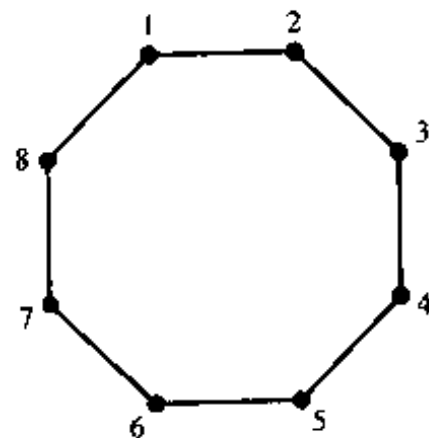
- Example. Let  $n$  be a positive integer, and let  $\rho_n$  denote the permutation of  $\{1, 2, \dots, n\}$

$$\rho_n = \begin{pmatrix} 1 & 2 & 3 & \dots & n-1 & n \\ 2 & 3 & 4 & \dots & n & 1 \end{pmatrix}$$

$$\rho_n^k = \begin{pmatrix} 1 & 2 & \dots & n-k & n-k+1 & \dots & n \\ k+1 & k+2 & \dots & n & 1 & \dots & k \end{pmatrix}$$

if  $r$  equals  $k \bmod n$ , then

$$\rho_n^r = \rho_n^k$$



- Example. Let  $n$  be a positive integer, and let  $\rho_n$  denote the permutation of  $\{1, 2, \dots, n\}$

$$\rho_n = \begin{pmatrix} 1 & 2 & 3 & \dots & n-1 & n \\ 2 & 3 & 4 & \dots & n & 1 \end{pmatrix}$$

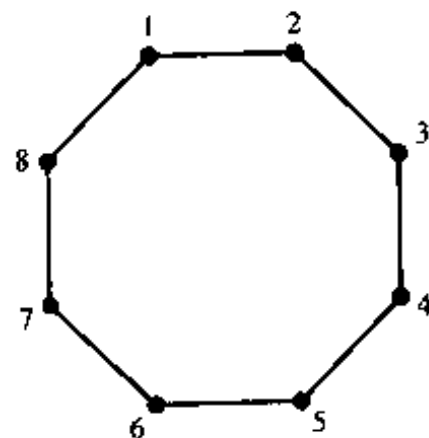
$$\rho_n^k = \begin{pmatrix} 1 & 2 & \dots & n-k & n-k+1 & \dots & n \\ k+1 & k+2 & \dots & n & 1 & \dots & k \end{pmatrix}$$

$$\rho_n^0 = \iota, \rho_n, \rho_n^2, \dots, \rho_n^{n-1}$$

$$\rho_n^{-1} = \rho_n^{n-1}$$

$$(\rho_n^k)^{-1} = \rho_n^{n-k}$$

$$C_n = \{\rho_n^0 = \iota, \rho_n, \rho_n^2, \dots, \rho_n^{n-1}\}$$



We may consider  $\rho_n$  as the rotation of the circle by an angle of  $360/n$  degrees.

The permutation is  $\rho_n^2$  then the rotation by  $2 \times (360/n)$  degree

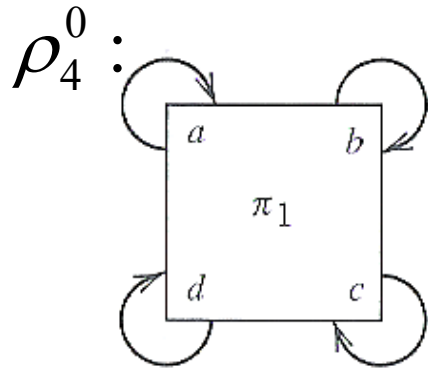
The permutation is  $\rho_n^k$  then the rotation by  $k \times (360/n)$  degree

Cycle group of order  $n$

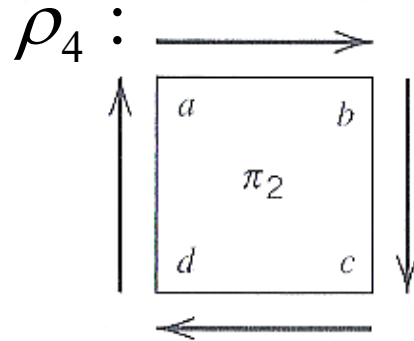
- Let  $\Omega$  be a geometrical figure. Each symmetry act as a permutation group  $G_C$  on its corner,  $G_E$  on the edge, and in the case of three-dimensional figure,  $G_F$  on the face.
  - A symmetry of  $\Omega$  is a (geometry) motion or congruence that brings the figure  $\Omega$  onto itself

# Rotations and Reflections (flips)

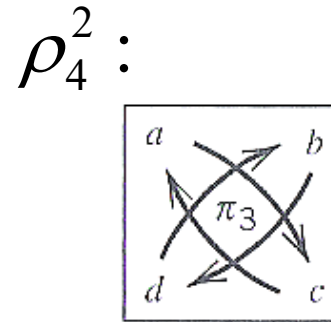
$\pi_1$ :  $0^\circ$  (zero rotation)  
(or  $360^\circ$ )



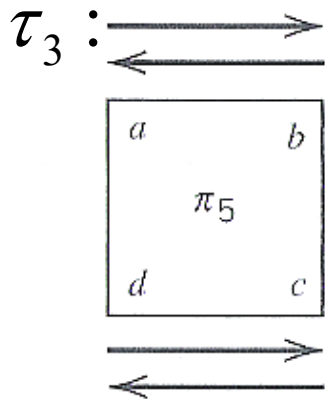
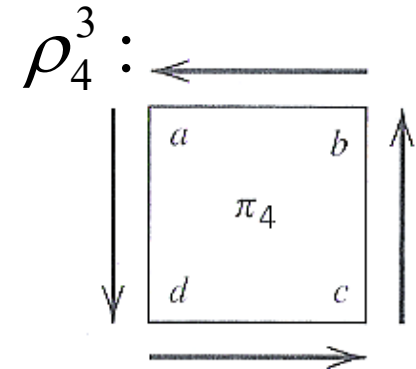
$\pi_2$ :  $90^\circ$



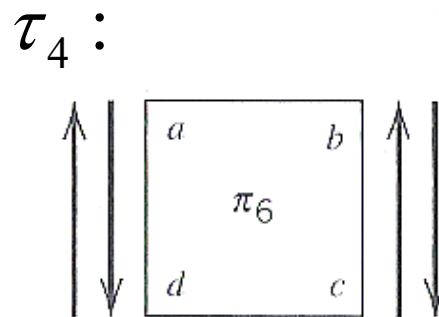
$\pi_3$ :  $180^\circ$



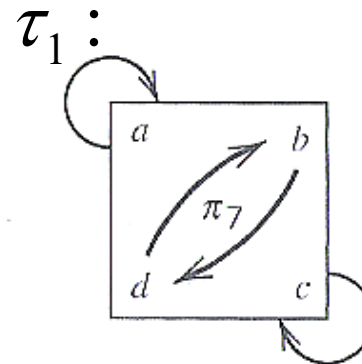
$\pi_4$ :  $270^\circ$



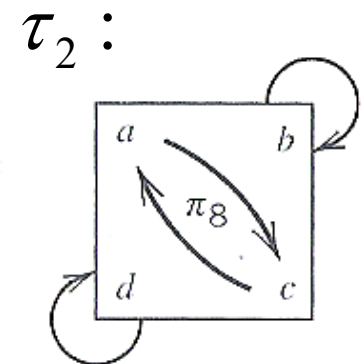
$\pi_5$ : reflection about the vertical axis



$\pi_6$ : reflection about the horizontal axis



$\pi_7$ : reflection about opposite corners  $a$  and  $c$

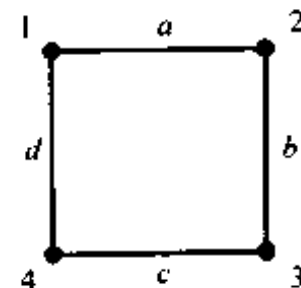


$\pi_8$ : reflection about opposite corners  $b$  and  $d$

- The rotation acting of 4 permutations

$$\rho_4^0 = \iota = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \rho_4 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix}$$

$$\rho_4^2 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \rho_4^3 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix}$$

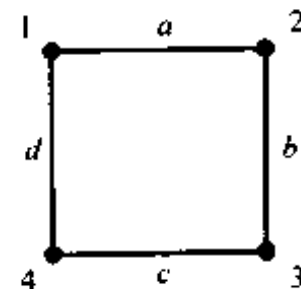


- The reflections acting of 4 permutations

$$\tau_1 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \tau_2 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 2 & 1 & 4 \end{pmatrix}$$

$$\tau_3 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \tau_4 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix}$$

$$G_C = \{\rho_4^0 = \iota, \rho_4, \rho_4^2, \rho_4^3, \tau_1, \tau_2, \tau_3, \tau_4\}$$

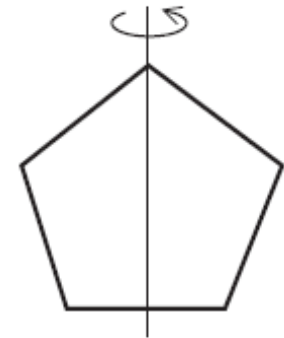


$$G_C = \{\rho_4^0 = \iota, \rho_4, \rho_4^2, \rho_4^3, \tau_1, \rho_4 \circ \tau_1, \rho_4^2 \circ \tau_1, \rho_4^3 \circ \tau_1\}$$

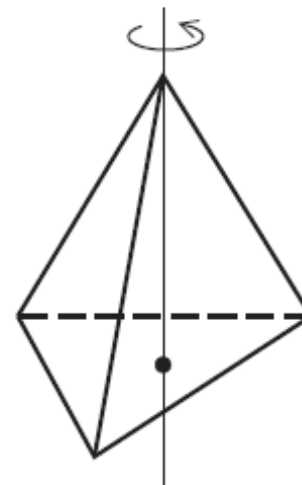
$$\tau_3 = \rho_4 \circ \tau_1, \tau_2 = \rho_4^2 \circ \tau_1, \tau_4 = \rho_4^3 \circ \tau_1$$

$$D_n = \{\rho_n^0 = \iota, \rho_n, \rho_n^2, \dots, \rho_n^{n-1}, \tau_1, \tau_2, \tau_3, \dots, \tau_n\}$$

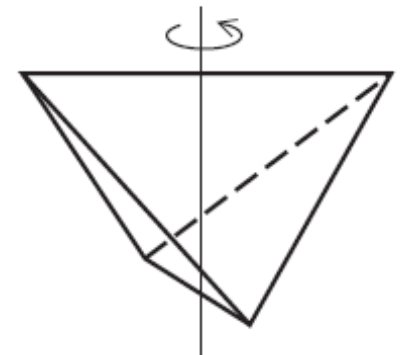
reflect about an axis of symmetry running from one corner to the middle of an opposite side  
There are five such reflections, for a total of 10 symmetries



**Figure 9.3** Symmetric reflection of a pentagon



(a)



(b)

**Figure 9.4** Symmetric revolutions of a tetrahedron

There are total of 1 ( $0^\circ$  motion) +  $4 \times 2$  + 3 = 12 symmetries

例. (10 阶二面体群) 如图 14-4 所示, 考察其顶点标以 1、2、3、4、5 的正五边形。它的 (角点) 对称群  $D_5$  包含 5 个旋转和 5 个反射, 5 个旋转为

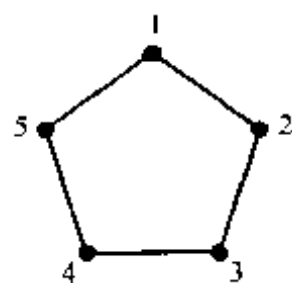


图 14 4

$$\rho_5^0 = \epsilon = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 3 & 4 & 5 \end{pmatrix} \quad \rho_5^1 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 4 & 5 & 1 \end{pmatrix}$$

$$\rho_5^2 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 1 & 2 \end{pmatrix} \quad \rho_5^3 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 5 & 1 & 2 & 3 \end{pmatrix}$$

$$\rho_5^4 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 1 & 2 & 3 & 4 \end{pmatrix}$$

将第  $i$  个角点与其对边的连线的反射记为  $\tau_i$  ( $i=1,2,3,4,5$ ), 则

$$\tau_1 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 5 & 4 & 3 & 2 \end{pmatrix} \quad \tau_2 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 2 & 1 & 5 & 4 \end{pmatrix}$$

$$\tau_3 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 4 & 3 & 2 & 1 \end{pmatrix} \quad \tau_4 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 1 & 5 & 4 & 3 \end{pmatrix}$$

$$\tau_5 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 3 & 2 & 1 & 5 \end{pmatrix}$$

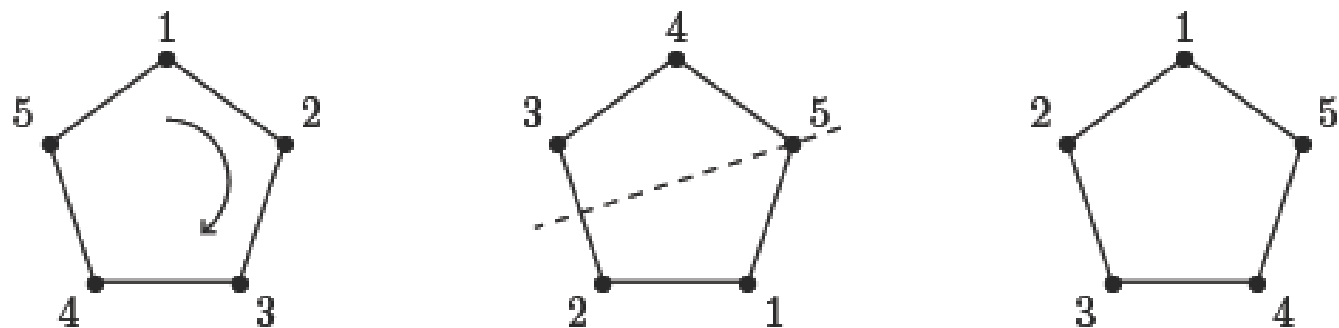


Figure 6.1.1 The rotation  $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 1 & 2 \end{pmatrix}$ .

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 2 & 1 & 5 & 4 \end{pmatrix} \circ \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 5 & 4 & 3 & 2 \end{pmatrix}.$$



Figure 6.1.2 The reflection  $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 2 & 1 & 5 & 4 \end{pmatrix}$ .



**Figure 6.1.3** The composition  $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 2 & 1 & 5 & 4 \end{pmatrix} \circ \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 5 & 4 & 3 & 2 \end{pmatrix}$ .

- The symmetries of a square are naturally characterized by the way they permute the corners of the square.
  - **cyclic permutation** or **cycle** are usually described the corner permutation
    - $x_1 \rightarrow x_2 \rightarrow x_3 \rightarrow \dots \rightarrow x_n \rightarrow x_1$
    - $(x_1 x_2 x_3 \dots x_n x_1)$

- Burnside's theorem

- Counting the number of nonequivalent coloring of a set  $X$  under the action of a group of permutation of  $X$

$$sN = c \quad \begin{array}{l} \text{: (number of symmetries) x (number of equivalence)} \\ \text{= (total number of coloring)} \end{array}$$

$$\therefore N = \frac{c}{s}$$

$$c = n!$$

$$s = n$$

$$N = \frac{n!}{n} = (n-1)!$$

Ex. The  $c=n!$  oriented seatings of  $n$  people around a round table. There are  $s=n$  cyclic rotations of the seatings, and each equivalence class consists of  $n$  seatings. Thus, the number of equivalence class (cyclicly nonequivalent seatings) is  $(n-1)!$

- Burnside's theorem

- Counting the number of nonequivalent coloring of a set X under the action of a group of permutation of X

$$sN = c$$

:(number of symmetries) x (number of equivalence)  
= (total number of coloring)

$$\therefore N = \frac{c}{s}$$

There are  $c=2^3=8$  "colorings" and  $s=3$  symmetries

$$N = \frac{3+1+1+1+1+1+1+3}{3} = \frac{12}{3} = 4$$

Need multiplicity correction!!

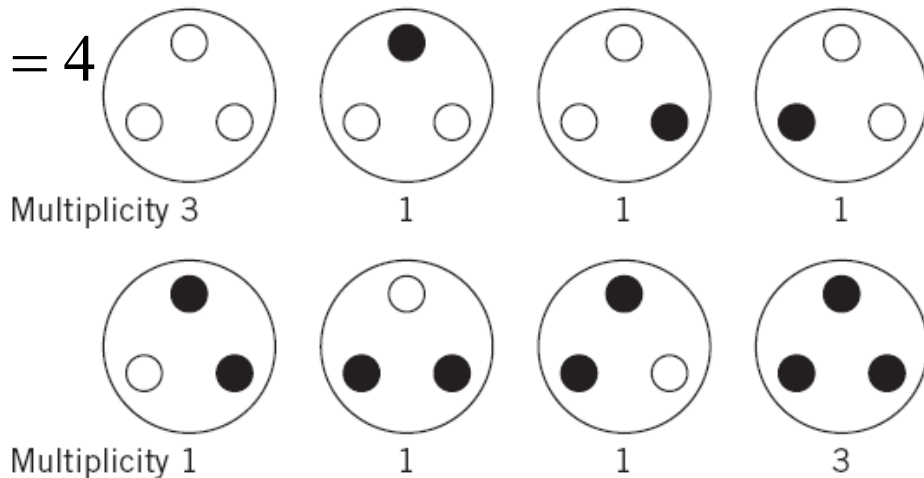


Figure 9.5

- Burnside's theorem

- Let  $G$  be a group of permutations of the set  $S$  (corner of a square). Let  $T$  be any collection of colorings of  $S$  (2-colorings of the corners) that is closed under  $G$ . Then the number  $N$  of equivalence class is

$$N = \frac{1}{|G|} \sum_{x \in T} \phi(x)$$

$$N = \frac{1}{|G|} \sum_{\pi \in G} \psi(\pi)$$

- where  $|G|$  is the number of permutations and  $\psi(\pi)$  is the number of colorings in  $T$  left fixed by  $\pi$ .

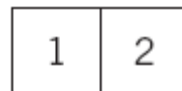
- Burnside's theorem

- Let  $G$  be a group of permutations of the set  $S$  (corner of a square). Let  $T$  be any collection of colorings of  $S$  (2-colorings of the corners) that is closed under  $G$ .

Then the number  $N$  of equivalence class is

$$\begin{aligned}
 (a) \quad N &= \frac{1}{|G|} \sum_{\pi \in G} \psi(\pi) & (b) \quad N &= \frac{1}{|G|} \sum_{\pi \in G} \psi(\pi) & (c) \quad N &= \frac{1}{|G|} \sum_{\pi \in G} \psi(\pi) \\
 &= \frac{1}{2} (2^2 + 2) = 3 & &= \frac{1}{2} (2^3 + 2 \times 2) = 6 & &= \frac{1}{2} (2^4 + 2 \times 2) = 10
 \end{aligned}$$

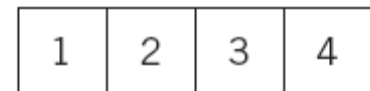
Ex. 2-colored batons



(a)



(b)



(c)

Figure 9.6

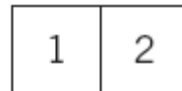
- Burnside's theorem

- Let  $G$  be a group of permutations of the set  $S$  (corner of a square). Let  $T$  be any collection of colorings of  $S$  (2-colorings of the corners) that is closed under  $G$ .

Then the number  $N$  of equivalence class is

$$\begin{aligned}
 (a) \quad N &= \frac{1}{|G|} \sum_{\pi \in G} \psi(\pi) & (b) \quad N &= \frac{1}{|G|} \sum_{\pi \in G} \psi(\pi) & (c) \quad N &= \frac{1}{|G|} \sum_{\pi \in G} \psi(\pi) \\
 &= \frac{1}{2} (3^2 + 3) = 6 & &= \frac{1}{2} (3^3 + 3^{\frac{3+1}{2}}) = 18 & &= \frac{1}{2} (3^4 + 3 \times 3) = 45
 \end{aligned}$$

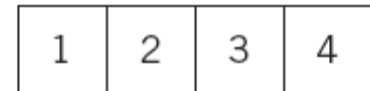
Ex. 3-colored batons



(a)



(b)



(c)

Figure 9.6

变即  $f * c = c$  的对偶  $(f, c)$  的个数。一种计数的方式是考察  $G$  中的每个  $f$ ，并计算  $f$  保持着色不变的着色数，然后相加所有的量。因  $C(f)$  是通过  $f$  保持着色不变的着色集，所以用这种方式计数得到

$$\sum_{f \in G} |C(f)| \quad (14-9)$$

另一种计数的方式是考察  $C$  中的每个  $c$ ，计算满足  $f * c = c$  的置换  $f$  的个数，然后相加所有的量。对每种着色  $c$ ，满足  $f * c = c$  的所有  $f$  的集合就是我们所称的  $c$  的稳定核  $G(c)$ 。因此，每个  $c$  对和的贡献是

$$|G(c)|$$

由推论 14.2.2 得

$$|G(c)| = \frac{|G|}{(\text{与 } c \text{ 等价的着色数})} \quad (14-10)$$

于是，用这种方式计数，得到

$$\sum_{c \in C} \frac{|G|}{(\text{与 } c \text{ 等价的着色数})} \quad (14-11)$$

但如果我们按等价类将着色归类，那么和式 (14-11) 可以简化。在同一等价类中，两种着色对和贡献了同样的量 (14-10)，因此，每个等价类的总贡献是  $|G|$ 。由于等价类的个数就是不等价的着色数  $N(G, C)$ ，所以，式 (14-11) 等于

$$N(G, C) \times |G| \quad (14-12)$$

使式 (14-9) 与 (14-12) 相等得到

$$\sum_{f \in G} |C(f)| = N(G, C) \times |G| \quad \square$$

解出  $N(G, C)$  即得 (14-8)。

例. (循环排列计数) 把  $n$  个不同的对象放在一个圆上, 问有多少种放法?

正如 13.1 节中所提示的那样, 问题相当于用  $n$  种不同的颜色对正  $n$  角形  $\Omega$  的角点进行着色, 答案应为  $\Omega$  的旋转群的不等价着色数。令  $\mathcal{C}$  是由对  $\Omega$  的  $n$  个顶点进行着色且每种颜色只出现一次的所有  $n!$  种方法所组成的集合, 则作用在  $\mathcal{C}$  上的循环群为<sup>①</sup>

$$C_n = \{\rho_n^0 = \epsilon, \rho_n, \rho_n^2, \dots, \rho_n^{n-1}\}$$

且循环排列的个数等于  $\mathcal{C}$  中不等价的着色数。 $C_n$  中的恒等置换  $\epsilon$  使  $\mathcal{C}$  中的  $n!$  种着色不变, 由于在  $\mathcal{C}$  的着色中每个顶点具有不同的颜色<sup>②</sup>, 所以  $\mathcal{C}$  中每个其他的置换均不保持  $\mathcal{C}$  中任何着色。因此, 根据定理 14.2.3 中的式 (14-8), 我们得到不等价的着色数为

$$N(C_n, \mathcal{C}) = \frac{1}{n} (n! + 0 + \dots + 0) = (n-1)!$$

□

例. (项链计数问题) 用  $n \geq 3$  种不同颜色的珠子镶成一个项链, 问有多少种方法?

这与上例的情形几乎相同, 只是项链可被翻转。现将置换群  $G$  取为一个正  $n$  角形的顶点对称群, 于是, 在这种情况下,  $G$  是阶为  $2n$  的二面体群  $D_n$ 。能使着色不变的唯一置换是恒等置换并且它使所有的  $n!$  种着色不变。所以, 由式 (14-8) 得到不等价的着色数即不同种项链数为

$$N(D_n, C) = \frac{1}{2n} (n! + 0 + \cdots + 0) = \frac{(n-1)!}{2} \quad \square$$

例：用红色与蓝色对一个正五角形的顶点进行着色，问有多少种不同的方法？

一个正五角形的对称群是二面体群

$$D_5 = \{\rho_5^0 = \epsilon, \rho_5, \rho_5^2, \rho_5^3, \rho_5^4, \tau_1, \tau_2, \tau_3, \tau_4, \tau_5\}$$

其中， $\tau_j$  是 13.1 节中所描述的关于连接顶点  $j$  与其对边中点连线的反射 ( $j = 1, 2, 3, 4, 5$ )。设  $C$  是关于正五角形顶点的所有  $2^5 = 32$  种着色的集合。对  $D_5$  中的各置换求出在其作用下保持不变的着色数，然后利用定理 14.2.3。单位元  $\epsilon$  使所有着色保持不变，其他四个旋转各自仅保持两种着色，即所有顶点为红色的着色和所有顶点为蓝色的着色不变，于是

$$|C(\rho_5^i)| = \begin{cases} 32, i=0 \\ 2, i=1, 2, 3, 4 \end{cases}$$

现考虑任意一个反射  $\tau_j$ ，比如  $\tau_1$ 。为了在  $\tau_1$  的作用下使着色保持不变，顶点 2 与 5 必须具有相同的颜色，且顶点 3 与 4 必须具有相同的颜色。所以，在  $\tau_1$  的作用下保持着色不变可通过如下方法获得：对顶点 1 选择一种颜色（两种选择）、对顶点 2 与 5 选择一种颜色（两种选择）、对顶点 3 与 4 选择一种颜色（两种选择）。因此，在  $\tau_1$  的作用下保持着色不变的着色数是  $2 \times 2 \times 2 = 8$ 。对其他各反射可进行类似计算，于是有

$$|C(\tau_j)| = 8 \quad \text{对每一个 } j = 1, 2, 3, 4, 5$$

因此，由式 (14-8) 得到不等价的着色数为

$$N(D_5, C) = \frac{1}{10}(32 + 2 + 2 + 2 + 2 + 8 + 8 + 8 + 8 + 8) = 8$$

□

例：用红色与蓝色对一个正五角形的顶点进行着色，问有多少种不同的方法？

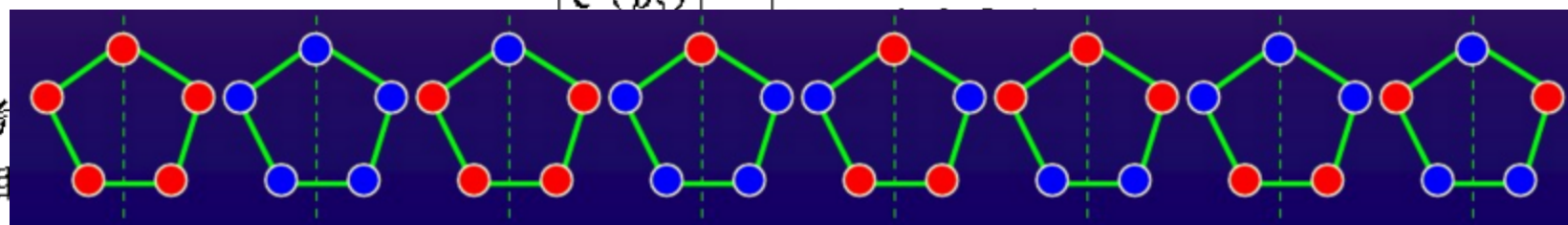
一个正五角形的对称群是二面体群

$$D_5 = \{\rho_5^0 = \epsilon, \rho_5, \rho_5^2, \rho_5^3, \rho_5^4, \tau_1, \tau_2, \tau_3, \tau_4, \tau_5\}$$

其中， $\tau_j$  是 13.1 节中所描述的关于连接顶点  $j$  与其对边中点连线的反射 ( $j = 1, 2, 3, 4, 5$ )。设  $\mathcal{C}$  是关于正五角形顶点的所有  $2^5 = 32$  种着色的集合。对  $D_5$  中的各置换求出在其作用下保持不变的着色数，然后利用定理 14.2.3。单位元  $\epsilon$  使所有着色保持不变，其他四个旋转各自仅保持两种着色，即所有顶点为红色的着色和所有顶点为蓝色的着色不变，于是

$$|C(\rho_5^i)| = \begin{cases} 32, & i = 0 \\ 2, & i = 1, 2, 3, 4 \end{cases}$$

现考  
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须具  
变可

通过如下方法获得：对顶点 1 选择一种颜色（两种选择）、对顶点 2 与 5 选择一种颜色（两种选择）、对顶点 3 与 4 选择一种颜色（两种选择）。因此，在  $\tau_1$  的作用下保持着色不变的着色数是  $2 \times 2 \times 2 = 8$ 。对其他各反射可进行类似计算，于是有

$$|C(\tau_j)| = 8 \quad \text{对每一个 } j = 1, 2, 3, 4, 5$$

因此，由式 (14-8) 得到不等价的着色数为

$$N(D_5, \mathcal{C}) = \frac{1}{10}(32 + 2 + 2 + 2 + 2 + 8 + 8 + 8 + 8 + 8) = 8$$

□

例：用红色、蓝色与绿色对一个正五角形的顶点进行着色，问有多少种不等价的方法？

参考上例，但现在的正五角形所有顶点的着色集  $C$  的元素个数是  $3^5 = 243$ ，单位元  $e$  使 243 种着色保持不变，其他每个旋转仅使 3 种着色保持不变。每个反射使  $3 \times 3 \times 3 = 27$  个着色保持不变。于是不等价的着色数为

$$N(D_5, C) = \frac{1}{10} (243 + 3 + 3 + 3 + 3 + 27 + 27 + 27 + 27 + 27) = 39$$

用  $p$  种颜色对一个正五角形的顶点进行着色，存在多少种不同的方法呢？推广以上的计算则得不等价的着色数应为

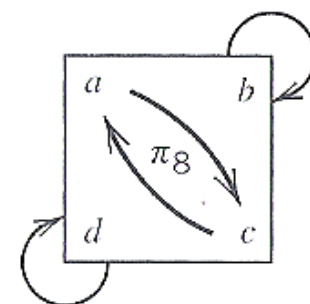
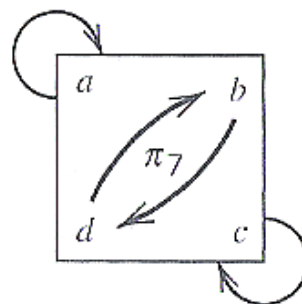
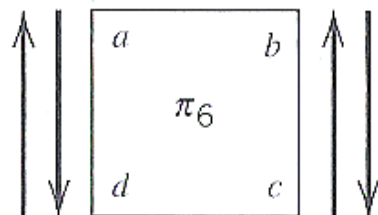
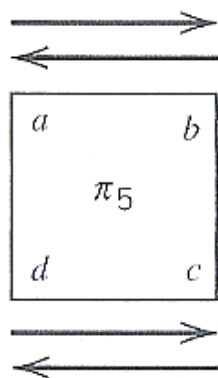
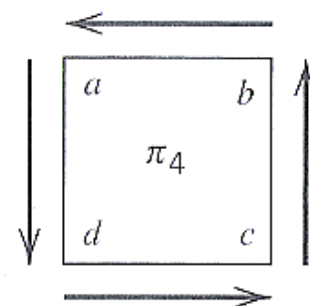
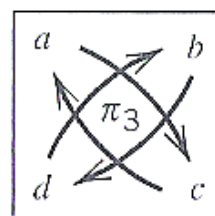
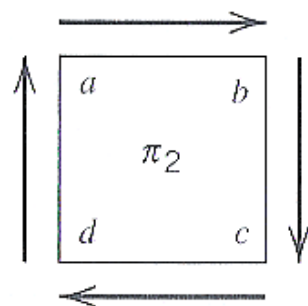
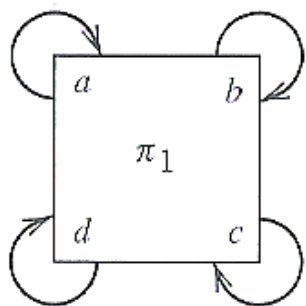
$$\frac{1}{10} (p^5 + 4 \times p + 5 \times p^3) = \frac{p(p^2 + 4)(p^2 + 1)}{10} \quad \square$$

- Example: 3-colored Necklaces
- Suppose a necklace can be made from beads of three colors – black, white, and red. How many different necklaces with  $n$  beads are there?

for  $n=3$ ,

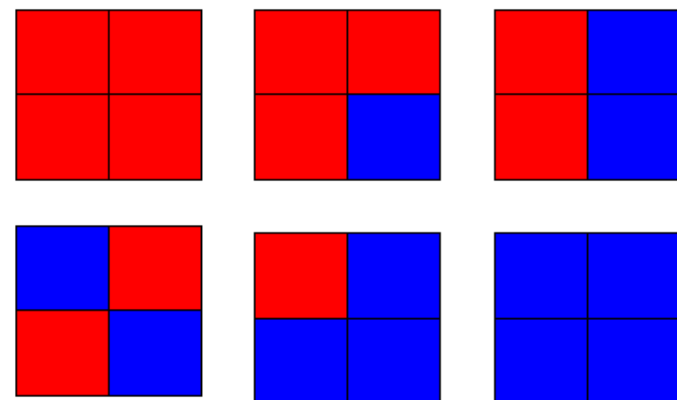
There are  $3^3 = 27$  3-colorings of a 3-bead necklace, and three rotations of  $0^\circ$ ,  $120^\circ$ ,  $240^\circ$ . The  $0^\circ$  rotation leaves all colorings fixed, and so  $\Psi(0^\circ) = 27$ . The  $120^\circ$  rotation cannot fix colorings in which some color occurs at only one corner. It follows that the  $120^\circ$  rotation fixes just the monochromatic colorings. Thus,  $\Psi(120^\circ) = 3$ . The  $240^\circ$  rotation is a reverse  $120^\circ$  rotation, and so  $\Psi(240^\circ) = 3$ . By formula (\*), we have

$$N = \frac{1}{3}(27 + 3 + 3) = 11 \blacksquare$$



$$N = \frac{1}{|G|} \sum_{\pi \in G} \psi(\pi)$$

$$= \frac{1}{8} (2^4 + 2 + 4 + 2 + 4 + 4 + 8 + 8) = \frac{48}{8} = 6$$



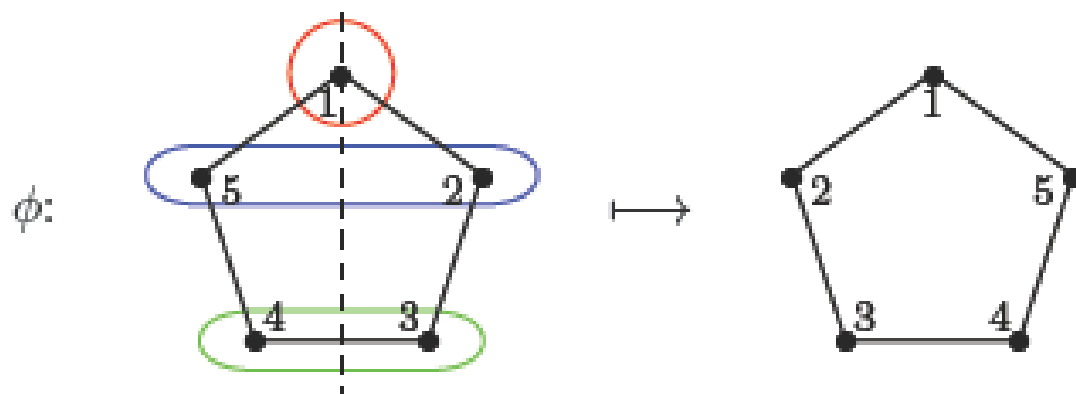
# • Pólya Counting Formula

正如上节中所看到的那样，在置换群  $G$  作用在着色集  $C$  的情况下，Burnside 定理之所以能成功地用于计数不等价的着色数，与它能够求出  $C$  中颜色不变的着色数  $|C(f)|$  有关，而  $C$  中的颜色不变是由  $G$  中的置换  $f$  保持的。通过考虑置换的循环结构，计算可变得容易简便。

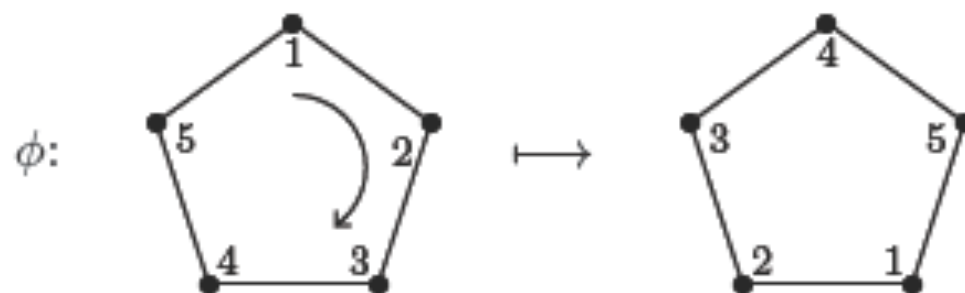
设  $f$  是  $X = \{1, 2, \dots, n\}$  的一个置换， $D_f = (X, A_f)$  是顶点集为  $X$  且弧集为

$$A_f = \{(i, f(i)) : i \in X\}$$

的有向图。该有向图有  $n$  个顶点与  $n$  个弧，各顶点的入度和出度等于 1。正如推论 11.8.8 所表明的那样，弧集  $A_f$  可以被划分为若干个有向圈，且每个顶点恰好只属于一个有向圈。理由很简单，因为从任意一个顶点  $j$  开始沿离开  $j$  的唯一弧继续向前并到达另一顶点  $k$ ，再对  $k$  继续重复该过程直至回到顶点  $j$ ，于是产生了一个有向圈。因为每个顶点的入度和出度均为 1，所以最终必然能到达我们的初始顶点  $j$ ，去掉如此获得的有向图的顶点和弧，直到  $D_f$  的所有顶点和弧取完为止，从而将  $D_f$  的顶点与弧全都划分成了有向圈。



**Figure 6.2.1** The cycles in a vertex permutation.



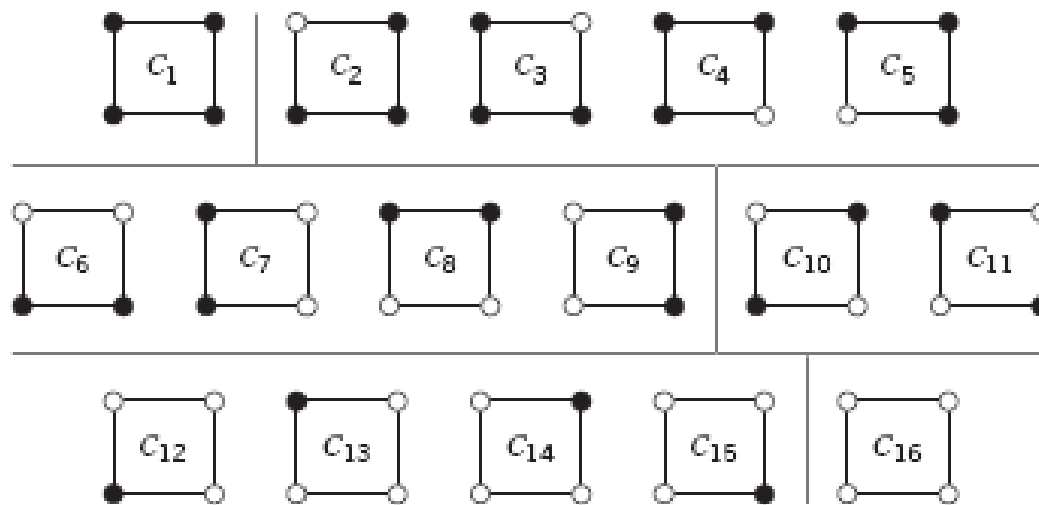
**Figure 6.2.2** The permutation  $(1, 3, 5, 2, 4)$  is a single cycle.

- Let  $S$  be a set of elements and  $G$  be a group of permutations of  $S$  that act to induce an equivalence relation on the colorings of  $S$ . The inventory of nonequivalent colorings of  $S$  using two colors is given by the generating function

$$P_G((b + w), (b^2 + w^2), (b^3 + w^3), \dots, (b^k + w^k))$$

| (i)<br><i>Motion</i><br>$\pi_i$ | (ii)<br><i>Colorings Left</i><br><i>Fixed by <math>\pi_i</math></i> | (iii)<br><i>Cycle Structure</i><br><i>Representation</i> |
|---------------------------------|---------------------------------------------------------------------|----------------------------------------------------------|
| $\pi_1$                         | 16—all colorings                                                    | $x_1^4$                                                  |
| $\pi_2$                         | 2— $C_1, C_{16}$                                                    | $x_4$                                                    |
| $\pi_3$                         | 2— $C_1, C_{10}, C_{11}, C_{16}$                                    | $x_2^2$                                                  |
| $\pi_4$                         | 2— $C_1, C_{16}$                                                    | $x_4$                                                    |
| $\pi_5$                         | 2— $C_1, C_6, C_8, C_{16}$                                          | $x_2^2$                                                  |
| $\pi_6$                         | 2— $C_1, C_7, C_9, C_{16}$                                          | $x_2^2$                                                  |
| $\pi_7$                         | 8— $C_1, C_2, C_4, C_{10}$<br>$C_{11}, C_{12}, C_{14}, C_{16}$      | $x_1^2 x_2$                                              |
| $\pi_8$                         | 8— $C_1, C_3, C_5, C_{10}$<br>$C_{11}, C_{13}, C_{15}, C_{16}$      | $x_1^2 x_2$                                              |

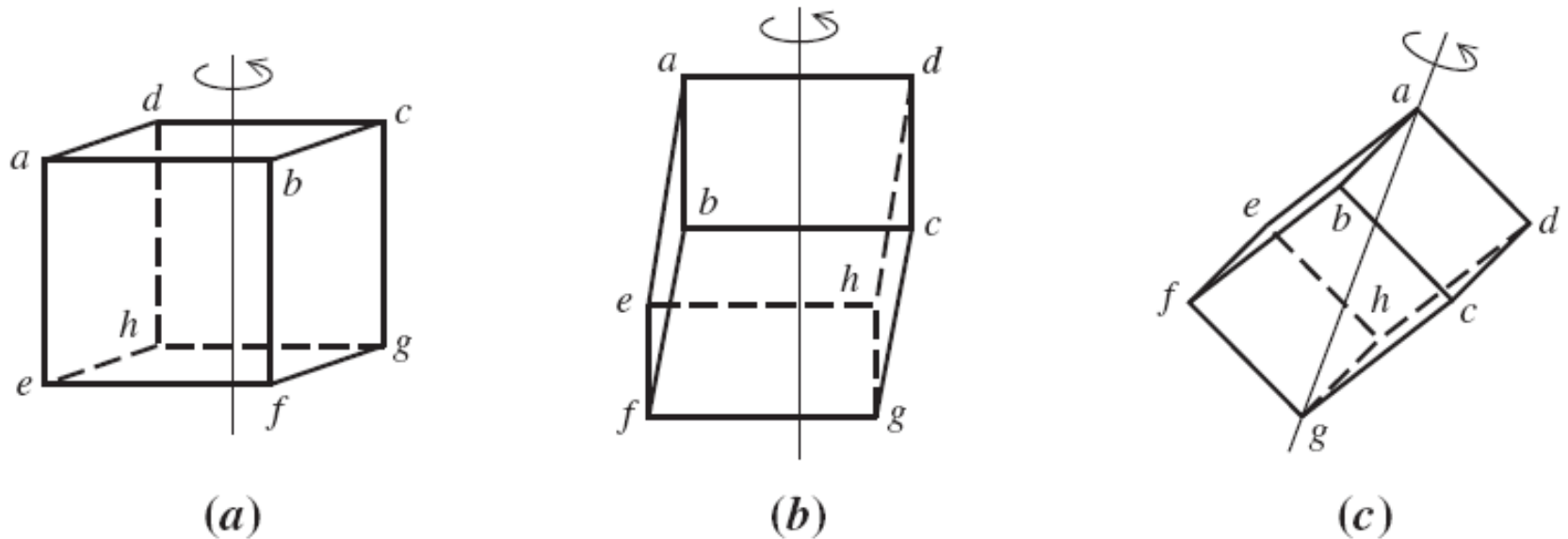
**Figure 9.7**



*Figure 9.1*

| (i)<br><i>Motion</i><br>$\pi_i$ | (ii)<br><i>Colorings Left</i><br><i>Fixed by <math>\pi_i</math></i> | (iii)<br><i>Cycle Structure</i><br><i>Representation</i> | (iv)<br><i>Inventory of Colorings</i><br><i>Left Fixed by <math>\pi_i</math></i> |
|---------------------------------|---------------------------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------------------------------|
| $\pi_1$                         | 16—all colorings                                                    | $x_1^4$                                                  | $(b + w)^4 = 1b^4 + 4b^3w + 6b^2w^2 + 4bw^3 + 1w^4$                              |
| $\pi_2$                         | 2— $C_1, C_{16}$                                                    | $x_4$                                                    | $(b^4 + w^4) = 1b^4 + 1w^4$                                                      |
| $\pi_3$                         | 2— $C_1, C_{10}, C_{11}, C_{16}$                                    | $x_2^2$                                                  | $(b^2 + w^2)^2 = 1b^4 + 2b^2w^2 + 1w^4$                                          |
| $\pi_4$                         | 2— $C_1, C_{16}$                                                    | $x_4$                                                    | $(b^4 + w^4) = 1b^4 + 1w^4$                                                      |
| $\pi_5$                         | 2— $C_1, C_6, C_8, C_{16}$                                          | $x_2^2$                                                  | $(b^2 + w^2)^2 = 1b^4 + 2b^2w^2 + 1w^4$                                          |
| $\pi_6$                         | 2— $C_1, C_7, C_9, C_{16}$                                          | $x_2^2$                                                  | $(b^2 + w^2)^2 = 1b^4 + 2b^2w^2 + 1w^4$                                          |
| $\pi_7$                         | 8— $C_1, C_2, C_4, C_{10}$<br>$C_{11}, C_{12}, C_{14}, C_{16}$      | $x_1^2 x_2$                                              | $(b + w)^2(b^2 + w^2) = 1b^4 + 2b^3w + 2b^2w^2 + 2bw^3 + 1w^4$                   |
| $\pi_8$                         | 8— $C_1, C_3, C_5, C_{10}$<br>$C_{11}, C_{13}, C_{15}, C_{16}$      | $x_1^2 x_2$                                              | $(b + w)^2(b^2 + w^2) = 1b^4 + 2b^3w + 2b^2w^2 + 2bw^3 + 1w^4$                   |

*Figure 9.9*



**Figure 9.10** Revolutions of the cube. (a) Revolution about opposite faces. (b) Revolution about opposite edges. (c) Revolution about opposite corners.